

ganisation working in the climate finance space, noted that adoption of renewable energy needs to accelerate. The year 2024 will mark the largest addition yet of non-fossil fuel energy capacity in India. “The 14.2 GW added in the first six months exceeds the 13.5 GW installed in 2023. But more is needed; annual installed capacity additions have to average 60GW by 2030,” the report said.

Besides, another problem that needs immediate attention is storage. That’s because the increase in green energy capacity does not automatically translate to a commensurate increase in green ener-

supply side push.

“While renewable energy capacity is increasing, the speed and scale need to accelerate. Further, grid modernisation, which includes smart grids with real-time monitoring and deploying flexible generation sources, will be crucial for RE grid integration,” says Garg.

INDUSTRIAL DECARBONISATION

One area that is key to the energy transition is the industrial sector, one of the major carbon emitters. Several top-listed companies like the Tata group firms and Reliance have announced their commitments and

pecially in hard-to-abate industries. Hygenco is setting up India’s first commercial long-term offtake green hydrogen plant with Jindal Stainless.

“Projects like these can go a long way in reducing the energy intensity of industries, especially as India continues to build significant infrastructure in the next two decades. Another application of green hydrogen is in refineries, where replacing grey hydrogen can reduce carbon emissions,” Bansal tells *BT*.

According to industry players, commercialisation of green hydrogen could also usher in a new era of clean energy-driven growth. “Overall, India’s clean energy transformation could open up new, exciting opportunities for India Inc. in the next 20 years and lay the foundation for a sustainable, secure, and resilient energy future as India turns 100,” says Aditya Lolla, Asia Programme Director at Ember, an independent energy think tank.

Technology will power this journey, but experts suggest that domestic solutions should be at the forefront. “India should lead the research and development of new technologies and promote domestic manufacturing to address energy security on one hand but also promote more job opportunities in energy and allied sectors,” says Garg. This will help in economic diversification and create alternative employment opportunities and will ensure that the transition happens in an equitable manner, she adds.

On a positive note, the Climake report found that corporate participation has grown significantly since 2020. In 2023, Indian corporate houses made up 47% of private investors in climate action.

FINANCE

That said, the transition will require a lot more investment. The International Energy Agency

gy generation. In 2023, non-fossil fuel energy accounted for 45% of electricity capacity but only 24% of generation. India needs to upgrade its grid to balance renewable energy’s variability.

The country must also upgrade transmission and distribution to integrate renewable energy. That last part alone requires an investment of \$30 billion. Till that is put in place, renewable energy generation will lag, highlighted the Climake report.

Vibhuti Garg, Director-South Asia at the Institute for Energy Economics and Financial Analysis (IEEFA), is of the view that the future of energy transition should include both demand side and

net-zero pathways.

This transition will require investment in low-carbon technologies such as carbon capture, utilisation, and storage, and ramping up work on alternative fuels such as green hydrogen. India launched the Green Hydrogen Mission with a capital outlay of ₹19,744 crore in 2022, keeping in mind industrial decarbonisation of hard-to-abate sectors like steel and cement. In addition to that, the Union Budget 2024-25 has proposed partnerships with private players to build small hybrid reactors for nuclear power.

Amit Bansal, Co-founder and CEO of green hydrogen producer Hygenco Green Energies, says green hydrogen can play a pivotal role, es-



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